

Do **not** write solutions on this page.

12. [Maximum mark: 22]

Consider the following homogenous differential equation

$$(x^2 + xy) \frac{dy}{dx} = x^2 + xy - 3y^2 \text{ where } x > 0 \text{ and } y > \frac{x}{2}.$$

It is given that $y = \frac{3}{2}$ when $x = 1$.

- (a) (i) Find the value of $\frac{dy}{dx}$ when $x = 1$.
 (ii) Use Euler's method with two equal steps to estimate the value of y when $x = 1.4$.
 (iii) Hence, state the concavity of the solution curve for $1 \leq x \leq 1.4$. You may assume that the concavity does not change in this interval. Give a reason for your answer. [6]

- (b) (i) Show that

$$(x^2 + xy) \frac{d^2y}{dx^2} = 2x + y - x \left(\frac{dy}{dx} \right)^2 - (x + 7y) \frac{dy}{dx}.$$

- (ii) Find the value of $\frac{d^2y}{dx^2}$ when $x = 1$. [6]

- (c) Determine constants $A, B \in \mathbb{R}$ such that $\frac{1+v}{1-4v^2} \equiv \frac{A}{1-2v} + \frac{B}{1+2v}$. [2]

- (d) By solving the differential equation $(x^2 + xy) \frac{dy}{dx} = x^2 + xy - 3y^2$ where $x > 0$, $y > \frac{x}{2}$ and $y = \frac{3}{2}$ when $x = 1$, show that $x^6(2y - x)^3 = 2(x + 2y)$. [8]

